INNOVATION BEAT

After losing $6 billion, Ginkgo Bioworks pivots to selling lab robots with AI

Ginkgo Bioworks CEO Jason Kelly, seen ringing the bell at the New York Stock Exchange in 2021, is pivoting the company to sell biotech lab robots with AI. COURTNEY CROW/NYSE

Jason Kelly, chief executive of Ginkgo Bioworks, spent a surprising amount of time on an earnings call with analysts last week talking about transportation, riding subways, and self-driving cars.

“If you’re sitting in the back of a Waymo, an autonomous car, you tell it where to go,” he explained. “Waymo’s AI takes over and it tells the autonomous car where to go.”

It’s not the usual topic for someone who runs a company focused on “synthetic biology,” the science Kelly and his cofounders helped develop at MIT to program cells as if they were software in an effort to develop new drugs, foods, and cosmetics.

But big changes are coming to Ginkgo as the Boston company struggles to find a profitable and growing business. Revenue has declined for three straight years and, since 2020, Ginkgo has racked up total net losses of $6 billion.

Ginkgo Bioworks' annual net loss

Source: US Securities and Exchange Commission

Partnering with biotech companies on synthetic biology research has been a dwindling business for several years, especially in Boston, since funding for life sciences dried up amid higher interest rates and competition from China.

Last year, the unit including Ginkgo’s synthetic biotech services brought in $133 million, a 24 percent decline from 2024 and less than one-eighth the amount Ginkgo forecasted it would have reached by now back when it went public in 2021 in one of the region’s biggest deals. Kelly has cut hundreds of jobs and Ginkgo’s stock price has plummeted more than 95 percent since the company hit a value of $20 billion in 2021.

Ginkgo Bioworks' Foundry revenue falls far below forecast

Chart: DANA GERBER/GLOBE STAFF • Source: US Securities and Exchange Commission

Employee count at Ginkgo Bioworks

Chart: DANA GERBER/GLOBE STAFF • Source: US Securities and Exchange Commission

So as part of a massive pivot, the company plans to close all of its traditional biotech labs, spin off its once-promising biosecurity business, and focus on building AI-driven robots for big drug companies and university labs (which explains why Kelly was talking about Waymo). Ginkgo already has deals with OpenAI and Google to help develop AI and cloud systems suitable for biotech researchers.

"The idea is to replace the lab bench," Kelly said in an interview. "What we would like to see, say five years from now, is that the majority of data generated at a place like a Merck or a Pfizer, or even in academic research institutions like an MIT, is generated with robotics rather than with scientists working at lab benches."

The race is already on to develop automated biotech research, with Lila Sciences in Cambridge talking about using AI not just to conduct experiments but to think them up, too.

Ginkgo has a 50-robot strong lab running at its offices in the Seaport and struck a $47 million deal in December to outfit the US Department of Energy's Pacific Northwest National Laboratory with almost 100 lab robots. The bots, dubbed Reconfigurable Automation Carts (or RACs), have wheels and robotic arms and can be combined in groups so they can do everything from preparing and cultivating thousands of samples to measuring the results with various instruments.

The company hopes to sell access to its automated lab over the cloud to smaller companies and individual scientists, while selling fully automated systems to large companies and universities, Kelly said.

"A scientist for as little as $39 can order an experiment on our setup in Boston and just get data back," Kelly said. The cloud business will provide immediate revenue this year, he said, while selling larger systems will take longer to ramp up.

Analysts and investors remain wary of Ginkgo, however, given its prior struggles.

Shares of Ginkgo, which had gained 17 percent in the first two months of the year, have dropped about 30 percent in the three days since Kelly announced the plan — mostly before the entire stock market dropped due to the war in Iran.

“Ginkgo is putting the pieces in place to establish itself as a winner in the AI drug discovery ecosystem,” analyst Matt Larew at William Blair & Co. wrote last week. “But as has been the case for several years, there needs to be tangible evidence of plans and promises translating to results for the stock to work.”

Kelly argues that lower-cost biotech research using Ginkgo’s robots could revive Boston’s life sciences sector. Automated labs can run nearly 24 hours a day, seven days a week, with minimal human staffing, he said.

“We’re the science capital of the country,” he said. “So if this trend keeps going, that would be a really good thing for us.”

🎭 How AI could be an economic miracle — or serial job killer. Read more from business columnist Larry Edelman.

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🔋 Form Energy to back Google data center with giant iron battery. Read more from tech columnist Hiawatha Bray.

🚓 Law enforcement software firmMultitude Insightsin Somerville raised$10 millionin a deal led by Primary Venture Partners.

💰 Semiconductor manufacturing tech startupMyrias Opticsin Amherst raised$2.1 millionin a deal led by MassVentures.

⛱️ Vacation rentals platform Breezeway in Boston received an unspecified investment from private equity firm Resurgens Technology Partners.

💵 Insurance tech firm Gradient AI in Boston received an unspecified investment from CIBC Innovation Banking.

🤝 Livechain, a subsidiary of Vyome Holdings in Cambridge, acquired HR data firm Humanyze, which started at the MIT Media Lab. Terms of the deal were not disclosed.

👋 Private equity firm Great Hill Partners in Boston hired Leland Lockhart as head of AI. Lockhart previously was executive director of artificial intelligence and machine learning at Vista Equity Partners.

☁️ Cloud storage company Wasabi Technologies added Zachary Smith, co-founder of Datum, to its board of directors.

🩺 Health care tech company Konovo in Watertown hired Jessica Santos as chief compliance & privacy officer. Santos previously was global compliance and quality director and data protection officer at Oracle Life Sciences.

Correction: Last week's newsletter left off a portion of new Flywire chief technology officer Patrick Blanc's prior experience. He was CTO of Visa's Cybersource and Value Added Services.

🤑 In his state of the union speech, President Trump said he was negotiating with major tech companies to pay a greater share of the energy costs associated with new data centers.

What We're Reading

The Pentagon's Favorite Tech Guy Is This Hawaiian Shirt-Wearing Founder (New York Times)

Inside Anthropic's Killer-Robot Dispute With the Pentagon (The Atlantic)

Block & Tackle: Job Cuts & the AI Narrative (Om Malik)

👋 *Thanks for reading. We'll be will be back next Tuesday.*

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